

IGCSE • Cambridge (CIE) • Maths

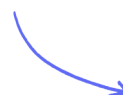
 40 mins  12 questions

Target Test

Created 29/05/26

Total Marks /40

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- 1** Ahmed sells different types of cake in his shop. The cost of each cake depends on its type and its size.

Every small cake costs \$ x and every large cake costs \$ $(2x + 1)$.

The number of small cherry cakes that can be bought for \$4 is the same as the number of large cherry cakes that can be bought for \$13.

Find the cost of a small cherry cake.

\$

(3 marks)

2 $2^p = \frac{1}{8^4}$

Find the value of p .

$p = \dots\dots\dots$

(2 marks)

- 3** Simplify.

$$2p - q - 3q - 5p$$

(2 marks)

- 4 (a)** t is inversely proportional to the square of $(x + 1)$.
When $x = 2$, $t = 5$.

Write t in terms of x .

$$t = \dots\dots\dots$$

(2 marks)

(b) When $t = 1.8$, find the positive value of x .

$$x = \dots\dots\dots$$

(2 marks)

5 $f(x) = 7 + 3x$.

Find $f^{-1}(x)$.

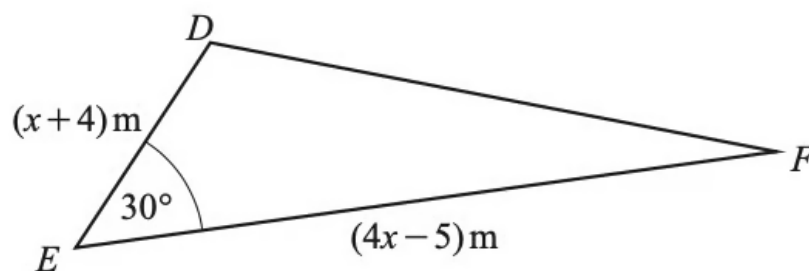
$$f^{-1}(x) = \dots\dots\dots$$

(2 marks)

6 Solve $10m^2 + 9m - 162 = 0$.

$$m = \dots\dots\dots \text{ or } m = \dots\dots\dots$$

(3 marks)



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The area of triangle DEF is 70 m^2 .

i) Show that $4x^2 + 11x - 300 = 0$.

[4]

ii) Use the quadratic formula to solve $4x^2 + 11x - 300 = 0$.

Show all your working and give your answers correct to 2 decimal places.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

iii) Find the length of DE .

$DE = \dots\dots\dots \text{ m}$ [1]

(9 marks)

- 8 Complete this statement with an expression in terms of m .

$$18m^3 + 9m^2 + 14m + 7 = (9m^2 + 7)(\dots\dots\dots)$$

(2 marks)

- 9 Make a the subject of the formula $p = \frac{3a + 5}{4 - a}$

(4 marks)

- 10 Solve $30 + 2x = 3(3 - 4x)$.

$$x = \dots\dots\dots$$

(3 marks)

- 11 $x^2 - 12x + a = (x + b)^2$

Find the value of a and the value of b .

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

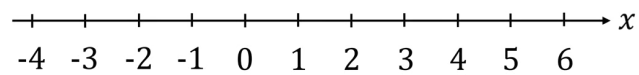
(2 marks)

12 (a) Solve the following linear inequality

$$4x + 5 \leq 17$$

(2 marks)

(b) Represent this inequality on the number line.



(2 marks)